Part I: Informed Consent

You are invited to participate in a research study conducted by [anonymous institution]. The purpose of the study is to examine potential coding security risks associated with the use of Next Edit Suggestion (NES) features in AI-integrated development environments (AI IDEs). You will be invited to fill out the online survey to give your opinion on AI inline suggestions integrated in the IDE. The survey would only take you about 5 minutes to complete, and you can choose to terminate the survey at any time without negative consequences. All information collected will remain strictly confidential. Individual details will not be disclosed or identifiable from this survey. If you have any questions about the research, please feel free to contact [anonymous instituition]. If you have questions about your rights as a research participant, please contact [anonymous instituition].

I understand the procedures described above and agree to participate in this study (tick box and proceed to Part II).

Part II: Questions

1. Current role:

    A. Professional developer / Software engineer

    B. Student

    D. Hobbyist

2. Programming experience:

    A. 0–3 years (Beginner)

    B. 3–5 years (Intermediate)

    C. 6–10 years (Advanced)

    D. More than 10 years (Professional)

3. Primary field or specialization:

    A. Computer Science / Software Engineering

    B. Cybersecurity / Information Security

C. Artificial Intelligence / Data Science

D. Other engineering disciplines (e.g., Electrical, Computer, Mechanical)

E. Other

4. Which of the following AI-integrated development environments or plugins have you used or are familiar with?

(Select all that apply.)

☐ GitHub Copilot

☐ Cursor

☐ Trae

☐ Zed

☐ Other

☐ None

5. Frequency of AI IDE usage: how often do you use AI-assisted IDEs or plugins during development?

A. Always (~90–100% sessions)

B. Frequently (~50–90% sessions)

C. Rarely (~10–50% sessions)

D. Never (<10%)

6. NES usage frequency: how often do you use NES features (e.g., automatic code completion or automated edits)?

A. Always (~90–100% sessions)

B. Frequently (~50–90% sessions)

C. Rarely (~10–50% sessions)

D. Never (<10%)

7. Reliance on NES: to what extent do you rely on NES features during development?

A. Low

B. Medium

C. High

8. Code review behavior: when NES suggests modifications to existing code, how often do you review the generated changes?

A. Always (~90–100%)

B. Frequently (~60-90%)

C. Sometimes (~30-60%)

D. Rarely (~1-30%)

E. Never (0%)

9. Factors influencing acceptance (select all that apply): under which conditions are you more likely to accept NES suggestions without detailed review?

☐ The code is well-formatted and stylistically consistent

☐ The suggested code is lengthy

☐ Nearby code includes test cases

☐ The code is non-critical (e.g., test or prototype code)

10. Exposure to unsafe suggestions: have you observed NES suggestions that introduce potentially unsafe coding practices?

A. Frequently (≥ 30%)

B. Occasionally (0-30%)

C. Never (0%)

11. Test artifacts in production code: have you observed NES suggesting test artifacts (e.g., hard-coded credentials, mock tokens) in production code?

A. Frequently (≥ 30%)

B. Occasionally (0-30%)

C. Never (0%)

12. Post-acceptance actions: after accepting an NES suggestion involving complex logic, what is your most common next step?

A. Continue coding without further review

B. Perform minor edits (e.g., formatting or renaming)

C. Verify functional correctness

D. Verify security properties

E. Execute the code or tests

HREC Reference Number: [anonymous reference number]